

Ratnesh Kumar Singh

Electronics & Communication Engineer | Networking & AI Automation

kumarsinghratnesh3@gmail.com

+91 70049 37129

Kolkata, West Bengal, India

LinkedIn: <https://tinyurl.com/2st86aht>

Portfolio: <https://my-portfolio-omega-liart-40.vercel.app>

Figma: <https://tinyurl.com/2c5nksav>

Github: <https://github.com/Ratnesh919>

SUMMARY

Motivated ECE graduate (2026) specializing in networking, IT infrastructure, and AI automation. Experienced in Python scripting, Figma UI/UX design, and embedded hardware development. Strong understanding of TCP/IP, routing, and V2X communication protocols. Eager to contribute to technical roles in systems integration, network engineering, or software development. Open to relocation across India.

SKILLS

Python • C • C++ • HTML • CSS • MATLAB • Arduino • Verilog • VS Code • TCP/IP • DNS • Routing & Switching • Cybersecurity • GIS (ArcGIS) • n8n • Google AI Studio • OpenAI API • AI Agents • Figma (UI/UX) • Visual Design • Canva • Adobe Photoshop • Adobe Lightroom • CapCut

PROFESSIONAL EXPERIENCE

Project Lead (Smart Antenna Prototype) | V2X Communication Research **Nov 2025 – Jul 2026**

- Designed and developed an adaptive smart antenna system prototype to improve wireless V2X communication in high-mobility environments.
- Implemented beamforming algorithms in Ansys HFSS software to dynamically direct signals toward moving vehicles, reducing interference.

AI Developer (Conversational Bot) | Python API Assistant Project **Aug 2024 – Nov 2024**

- Developed a modular conversational assistant utilizing Python scripting and REST API integrations for real-time data retrieval.
- Designed clean client-server communication protocols to handle conversational history, query parsing, and error safety.

IoT Hardware Developer | Smart Parking System **May 2024 – Jul 2024**

- Created an IoT smart parking prototype using Arduino microcontrollers and ultrasonic sensors for real-time occupancy tracking.
- Programmed sensor polling algorithms and integrated wireless communication shields to transmit telemetry updates.

UI/UX & Web Developer | Figma Prototypes & Portfolio **Jan 2024 – Apr 2024**

- Designed multi-screen UI layouts and interactive wireframes in Figma, enforcing WCAG accessibility guidelines.
- Developed a highly responsive portfolio website using HTML, CSS, and deployed the live build on Vercel.

EDUCATION

B.Tech in Electronics & Communication Engineering **2022 – 2026**

Swami Vivekananda Institute of Science & Technology | Board/University: MAKAUT (W.B.U.T.)

12th Standard - Science (PCM) **2018 – 2020**

P.B.S College, Bihar | Board/University: B.S.E.B

10th Standard - Science **2017 – 2018**

Vidyanjali High School (H.S.) | Board/University: I.G.C.S.E

ACHIEVEMENTS & CERTIFICATIONS

- GIS (Geographic Info. Systems):** Completed a 2-week training course on geospatial mapping, spatial data annotation, and data tagging.
- Python Programming:** Completed a 1-month certification course focused on python scripting, algorithms, and automation.
- C Language Programming:** Completed a 1-month certification course on data structures, pointers, and memory diagnostics.
- EV Service Technician:** Completed a 1-month program on electric vehicle electronics, system diagnostics, and hardware testing.
- Extra-Curriculars:** Participated in university workshops on HAM Radio signal monitoring, BSNL telecom systems, and Cybersecurity awareness.

LANGUAGES & INTERESTS

Languages: English (Professional), Hindi (Native), Bengali (Conversational)

Interests: Video gaming (analysis of mechanics/genres), photography, sketching, image/video editing, and tracking tech trends